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CHEMISTRY 1
0715

BAMENDA ARCHDIOCESAN EXAMINATION BOARD (BAEBOC)
MOCK G.C.E. EXAMINATION

MARCH 2025 ADVANCED LEVEL

Centre N°. & Name	
Candidate N°	GTPA026.
Candidate Name	

0715 Chemistry: Multiple - Choice Questions Paper 1

Time: 1 hour 30 mins.

INSTRUCTIONS TO CANDIDATES

Read the following instructions carefully before you start answering the questions in this paper. Make sure you have a soft HB pencil and an eraser for this examination.

1. USE A SOFT HB PENCIL THROUGHOUT THIS EXAMINATION
2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Before the examination begins.

3. Check that this question booklet is headed "Advanced Level Chemistry 0715"
4. Insert the information required in the spaces above.
5. Without opening the booklet, pull out your answer sheet carefully from inside the front cover of this booklet. Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instructions.
6. Insert the information required in the spaces provided on the answer sheet using your HB pencil: Candidate Name, Centre Number and Name, Candidate Number, Subject Code Number and Paper Number.

HOW TO ANSWER THE QUESTIONS IN THE EXAMINATION.

7. Answer all questions in this section.
8. Each question has FOUR suggested answers: A, B, C AND D. Decide on which answer is correct. Find the number of the question on the answer sheet and draw a thick horizontal line across the letter to join the square brackets for the answer you have chosen.
   For example, if C is your correct answer, mark C as shown below:
   [A] [B] [C] [D]
9. Mark only one answer for each question. If you mark more than one answer, you will score zero for that question. If you change your mind about an answer, erase the first mark carefully, then mark your new answer.
10. Avoid spending too much time on any one question. If you find a question difficult, move on to the next question. You can come back to that question later.
11. Do all rough work in this booklet, using where necessary, the blank spaces in the question booklet.
12. Mobile phones are not allowed in the examination room.

TURN OVER

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Questions 1-37 each of the question have four suggested answers.; Complete the statements with the correct response.

1) The basic hydroxide that will not produce hydrogen gas when heated
   A) sodium hydroxide
   B) lithium hydroxide
   C) calcium hydroxide
   D) magnesium hydroxide

2) The pH of aqueous solution of benzoic acid of concentration
   1.0Moldm^{-3} and Ka = 6.0 x 10^{-4} moldm^{-3}
   A) 3.22
   B) 4.22
   C) 2.11
   D) 1.60

3) Given the total bond enthalpy term in KJmol^{-1} for the following substances; cyclohexene = +6471, hydrogen =+436, cyclohexane =+7032. What would be the enthalpy change for the reaction



   A) -997
   B) -125
   C) +997
   D) +125

4) Which of the following techniques would be most appropriate in determining the arrangement of ions in a crystal
   A) nuclear magnetic resonances
   B) x-rays diffraction
   C) mass spectrometer
   D) infrared spectrometer

5) Given the data:

$Cu^{2+}_{(aq)} + e' \longrightarrow Cu^{+}_{(aq)}$ $\quad E^{\theta} = +0.15v$

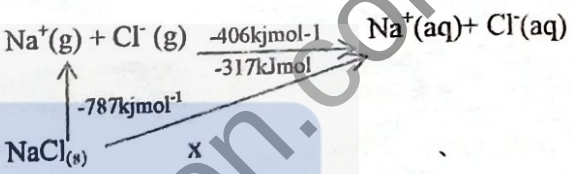
$Cu^{2+}_{(aq)} + 2e' \longrightarrow Cu_{(s)}$ $\quad E^{\theta} = +0.34v$

from the data above it could be deduce that:
   A) the e.m.f of the cell formed by the two electrodes is -0.19v
   B) the cell diagram is $Cu_{(s)}/Cu^{2+}_{(aq)} // Cu^{2+}_{(aq)}, Cu^{+}_{(aq)}/Pt_{(s)}$
   C) the mass of $Cu_{(s)}$ decreases as the cell operates
   D) $Cu^{+}_{(aq)}$ will disproportionate in aqueous solution

6) The concentration in Moldm^{-3} when 5.21g of sulphamic acid (NH_2SO_3H) is dissolve in 250cm^3 of solution is (RAM N=14, S=32, O=16 ,H=1)
   A) 0.05
   B) 0.24
   C) 0.10
   D) 0.21

7) A positive deviations from Raoult's law could be obtained from a solution of:
   A) methane /water
   B) ethanol/water
   C) methanoic acid/water
   D) nitric acid/water

8) Consider the energy cycle below for sodium chloride



   The value for the enthalpy change represented by X is
   A) -1293
   B) -1193
   C) -1510
   D) -713

9) A solution of 0.05MK_2CO_3 solution was used for a titration in which 25cm^3 K_2CO_3 solution required 24.6cm^3 of hydrochloric acid. The concentration of hydrochloric acid used is:
   A) 0.005M
   B) 0.05M
   C) 1.02M
   D) 0.102M

10) Which of the following species could function as a chealating ligand:
   A) $C_2O_4^{2-}$
   B) ClO_3^-
   C) NH_3
   D) H_2O

11) A liquid like water will boil when
   A) it vapour pressure is equal to saturated vapour pressure
   B) it vapour pressure is greater than atmospheric pressure
   C) it vapour pressure is equal to the atmospheric pressure
   D) it saturated vapour pressure is equal to the atmospheric pressure

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12) The shape of the complex compound $NaCoCl_4$ is:
A) linear
B) square planar
C) tetrahedral
D) octahedral

13) For the group iv element C - Pb , choose the correct statement ;
A) The stability of the +4 oxidation state increase down the group
B) All the tetrachlorides hydrolyses in water
C) PbO is amphoteric while PbO_2 is acidic
D) Tin (II) compounds is slightly less stable than tin (IV) compoounds

14) The compound that will produce a yellow precipitate with 2,4 – dinitrophenylhydrazine reagent and also with alkaline iodine solution
A) Butan-2-ol
b) Butanal
C) Butanone
D) Pentan-3-one

15) One mole of N_2O_4 was allowed to dissociation and reach equilibrium at 400°C. At equilibrium the partial pressure of N_2O_4 was found to be o.15atms. The kp for the reaction $N_2O_{4(g)} \rightleftharpoons 2NO_2$ is 48 atms .The partial pressure of NO_2 in atms for the equilibrium mixture at 400^0C is:
A) 2.68
B) 7.20
C) 0.56
D) 1.08

16) Which of the following results from the addition of hydrochloric acid onto nbutylamine (l-aminobutane).
A) $C_4H_7NH_2$
B) $C_4H_9NH_3^+$
C) $C_4H_9NH_2^+$
D) $C_4H_7NH_3^+$.

17).Which of the following indicators is suitable for the standardization of silver nitrate solution using sodium chloride solution.
A) phenolphthalein
B) methyl orange
C) chromate
D) starch

18) The abnomalous behavior of lithium in group I is due to
A) its high electronegativity
B) its small size
C) its charge density
D) its position in P.T

19) An anion of sulphur with formula S^{2-} has the electronic configuration
A) $1s^2 2s^2 2p^6 3s^2 3p^5$
B) $1s^2 2s^2 2p^6 3s^2 3p^4$
C) $1s^2 2s^2 2p^6 3s^2 3p^3$
D) $1s^2 2s^2 2p^6 3s^2 3p^6$

20) How many sodium ions surround each chloride ion in a unit cell of sodium chloride crystal?
A) 2
B) 6
C) 4
D) 8

21) Which of the following compound will give a white precipitate immediately with concentration hydrochloric acid/zinc chloride reagent?
A) $CH_3CH_2CH_2OH$
B) $CH_3CHOHCH_3$
C) $CH_3CH(CH_3)CH_2OH$
D) $(CH_3)_3COH$

22) Given the half cells Sn^{+4}/Sn^{+2} E=+O. 15v Fe^{+3}/Fe^{+2} E = +O.77v, the reaction is $Sn^{+2} + Fe^{+3} \longleftrightarrow Sn^{+4} + Fe^{+2}$. The correct statement below is
A) The cell diagram is $Pt/Sn^{+2},Sn^{+4} // Fe^{+3}, Fe^{+2}/Pt$
B) The spontaneous reaction is from right to the left
C) Sn is the species that has been reduced
D) Fe is the species that has been oxidized

23) The coordination number of cobalt in the complex $[Co(en)_2Br_2]$ is
A) 6
B) 3
C) 4
D) 5

24) A lone pair of electrons in a molecule is:
A) closer to the atom than a bonded electron
B) has a strong repulsive force on the molecule and reduce the bond angle
C) Distort the shape of the molecule and reduce the bond angle
D) Distort the shape of the molecule and increase the bond angle

25) The bond angle of nitric acid (HNO_3) is
A) 109.5°
B) 120°
C) 180°
D) 104.5°

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26) Electronegativity is defined as;
A) The ability of an element to accept an electron
B) The energy required by an element to give out electron
C) The energy required by an element to accept an electron
D) The ability of an element to donate electrons

27) What account for phosphorus to form two chlorides while nitrogen could form only one chloride?
A) phosphorus has a larger size than nitrogen
B) phosphorus exist as a solid white nitrogen is a gas
C) phosphorus can expand its octet and not nitrogen
D) Nitrogen has only one lone pair of electrons

28) Given the following standard enthalpies of combustion in $KJmol^{-1}$, the enthalpy of formation of methane (CH_4) in $KJmol^{-1}$ is, (hydrogen =-286 Carbon =-394 , methane=-890)
A) +76
B) -76
C) +230
D) -230

29) For the reaction A+B $\longleftrightarrow$ C, the following data was obtained

[A]moldm^{-3}	[B]moldm^{-3}	Related rate
0.1	0.1	1
0.2	0.1	4
0.1	0.2	2

The reaction is
A) First order with respect to A
B) second order with respect to B
C) second order with respect to A
D) second order overall

30) Ozone can react with nitrogen dioxide as follows:
$O_3(g) + 2NO_2(g) \longrightarrow N_2O_5(g) + O_2(g)$
The mechanism of the reaction is
$O_{3(g)} + NO_2 \longrightarrow NO_{3(g)} + O_g$ slow step
$NO_{2(g)} + NO_{3(g)} \longrightarrow N_2O_5$
The rate is:
A) Rate=$[No_3][No_3]$
B) Rate= $k[NO_2][NO_3]$
C) Rate=$K[O_3][NO_2]$
D) Rate=$K[N_2O_5]$

31) Which of the following pairs of liquid mixtures will obey Raoult's law?
A) water/ethanol
B) methanol/ethanol
C) propanone/trichloromethane
D) propanone/hexane

32) The element which could show diagonal relationship with the element aluminums
A) lithium
B) magnesium
C) beryllium
D) silicon

33) The salt that will produce an apple flame over a burner flame
A) Strontium nitrate
B) barium chloride
C) potassium chloride
D) copper nitrate

34) A group(iv) element whose divalent ion readily decolorizes acidify potassium per`manganate solution
A) Silicon
B) Germanium
C) Lead
D) Tin

35) The compounds copper (I) are colourless because
A) copper has lost one electron
B) the 3d orbital is partially filled
C) the 3d orbital is filled preventing electron transfer
D) it is the last transition metal

36) The properties of the nucleus that lead to it discovery is (are)
A) large empty space
B) small size
C) positive charge and small size
D) position in the atom

37) The monoxide that is not readily oxidize to the dioxide when heated
A) Carbon monoxide
B) lead(ii)oxide
C) silicon monoxide
D) germanium(II)oxide

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For Questions 38-50 use the following options

A) if 1,2and 3 are correct
B) if 1and 2 are correct
C) if 2and3 are correct
D) if 3 only is correct

A)	B)	C)	D)
1,2,3 correct	1,2 correct	2,3 correct	3 correct

38) what account for the high stability and high melting point of amino ethanoic acid (NH_2CH_2COOH)
1) it exist as a zwitterion in neutral medium
2) it exist as a dipolar ion and behaves as an ionic compound.
3) it is an acid in acid medium and a base in basic medium

39) The pH of ethanoic acid solution of concentron 3 x 10^{-5}moldm^{-3} (Ka = 1.8 x 10^{-5}) is:
1) 4.7
2) 4.26
3) 4.63

40) The compound of magnesium carbonate decompose on heat while sodium carbonate does not; this is due to
1) small size of the sodium and carbonate ion
2) Small size of magnesium ion and large size of the carbonate ion
3) large size of the carbonate ion which decompose to oxide

41) consider the equilibrium reaction
$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$ when the pressure is increase
1) The equilibrium shift to the right
2) the equilibrium shift to the left
3) There is no change in the equilibrium position

42) Iodine solution is formed when iodine crystal dissolve in potassium iodide. What account for this?
1) both substance have van der Waals forces
2) both substance have similar bond type
3) iodine forms a soluble complex with potassium iodide

43) Which of the following compounds could dimerise in the gaseous phase
1) methanol(CH_3OH)
2) ethanoic acid (CH_3COOH)
3) Methanoic acid (CHO_2H)

44) Which of the following steps is(are) involved in the reaction of propane with chlorine
1) heterolytic cleavage of the chlorine molecule
2) termination reaction form chloropropane
3) formation of free radicals

45) The bond type(s) found in sodium borohydride($NaBH_4$)
1) ionic bond
2) covalent bond
3) hydrogen bond

From question 46-50 use the following to answer the questions

A	TRUE	TRUE	Second statement correctly explains the first
B	True	True	Second statement is not a correct explanation of the first
C	True	False	
D	false	True	

46) The nitrate of group I decompose on heating to the corresponding oxides NO_2 and O_2	The thermal stability of group I nitrates increase down the group
47) Barium sulphate is insoluble while calcium sulphate is soluble in water.	The lattice energy of barium sulphate is too high and cannot be compensated for by hydration energy
48) The radius of the fluoride ion is much greater than that of fluorine atom	When halogens form anions the added electron complete the outer most shell
49) The oxalate ion $C_2O_4^{2-}$ is a bidentate ligand	In the oxalate ion the donor atom is carbon
50) phenol will form a yellow precipitate when it reacts with cold benzene diazonium ions in solution	This is a coupling reaction.

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